

Scale or Rotation? Decomposing Neural Tangent Kernel Movement During Grokking

Group G2

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[TODO: Need to verify all citations by hand]

1 Topic

Grokking is delayed generalisation: on modular addition the training loss falls early, the test loss stays at chance for orders of magnitude more steps, then collapses abruptly (Power et al., 2022). The leading dynamical explanation is a transition from *lazy* (kernel) to *rich* (feature-learning) training: memorisation is what a linearised model can do, and generalisation requires the a model $f(x; \theta_t)$ ’s neural tangent kernel (NTK) $\Theta_t(x, x') = \langle \nabla_{\theta} f(x; \theta_t), \nabla_{\theta} f(x'; \theta_t) \rangle$ to move away from its initial Θ_0 (Jacot, Gabriel, and Hongler, 2018; Kumar et al., 2024).

“The kernel moves” is two claims fused into one: a kernel can move in overall *scale* S_t (all eigenvalues rescale, eigenbasis fixed) or in *rotation* R_t (the eigenbasis rotates), and the standard summary statistic does not separate them. Defining $S_t = \log(\|K_t\|_F / \|K_0\|_F)$ and $R_t = 1 - \langle \hat{K}_t, \hat{K}_0 \rangle_F$, given the normalised empirical neural tangent kernel as $\hat{K} = K / \|K\|_F$, we may decompose the standard metric of kernel change as,

$$\|K_t - K_0\|_F^2 / \|K_0\|_F^2 = e^{2S_t} + 1 - 2e^{S_t}(1 - R_t), \quad (1)$$

so relative kernel movement accounts for both, and there is no way to derive which type of movement occurred from it. This is load-bearing. A network is k -homogeneous if scaling every parameter by c scales its output by c^k , with a bias-free ReLU MLP having k equal to its number of layers. Lewkowycz and Gur-Ari (2020, Thms. 1–2) show that an infinitely wide k -homogeneous network with L_2 regularisation coefficient λ and MSE has the theoretical neural tangent kernel $\Theta_t = e^{-2(k-1)\lambda t} \Theta_0$ so $S_t = -2(k-1)\lambda t$ with $R_t \equiv 0$ which is pure scaling with zero feature learning. Yet this is the theorem Kumar et al. (2024, App. 13) invoke to argue that weight decay moves a network out of the lazy regime, hence $t_{\text{grok}} \propto 1/\lambda$, an argument they label speculative and support with no measurement of kernel change. Conversely, a model that stays in the kernel regime provably needs a constant training fraction of the p^2 possible sums (Mohamadi et al., 2024), so at low training fractions rotation is not optional.

Research question. What sets grokking time t_{grok} — kernel rotation, kernel or weight scale, or the weight-decay rate itself? Three 2026 accounts disagree (§2), and every observable they use is a magnitude: unmoving if the kernel’s eigenbasis rotates, but moving if it merely rescales.

Hypothesis H (rotation clock). Write K_t for the empirical NTK on a fixed probe set of n inputs, Y for the one-hot label matrix (so $YY^\top$ records which inputs share an answer), $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ for the centring matrix, and A_t for the cosine between HK_tH and $HYY^\top H$ — the centred kernel–target alignment (Cortes, Mohri, and Rostamizadeh, 2012), which is unchanged by any rescaling of K_t and so responds only to rotation. H is that generalisation begins when A_t first crosses a threshold A_* ; that A_* is common to all cells of the sweep $(\alpha, \eta\lambda, N, p)$; and that, given A_t , t_{grok} carries no further dependence on S_t .

Discriminating prediction. In the exact evolution $\dot{\Theta}_t = -2(k-1)\lambda\Theta_t + T_t + T_t^\top$ (Lewkowycz and Gur-Ari, 2020, Eq. S5) the weight-decay term is a scalar multiple of Θ_t and only rescales the kernel, which leaves A_t unchanged. Alignment can move only through T_t , and T_t vanishes at infinite width (Jacot, Gabriel, and Hongler, 2018; Lewkowycz and Gur-Ari, 2020), so rotation is a finite-width effect: the wider the network, the slower it turns. Under H the target A_* is fixed, so t_{grok} is a distance divided by a rate — it should grow with N and be flat in $\eta\lambda$. Fitting $\log t_{\text{grok}} = c + a \log N + b \log(\eta\lambda)$ on a crossed grid, H gives $a > 0$, $b \approx 0$ ($a = 1$ if the correction is $O(1/N)$); a weight-decay clock gives $a \approx 0$, $b \approx -1$; both nonzero means two clocks, the slower binding. We estimate a, b rather than assuming either.

2 Significance

The 2026 dispute. Truong et al. (2026) show causally that the weight norm sets the timescale: clamping $\|W\|$ at $\rho\|W\|_c$ makes t_{grok} grow exponentially in ρ , with $\|W\|_c \propto p^{0.38}$ and an exponent of ≈ 7.5 for an MLP against ≈ 15 for unnormalised attention — the form of the law transfers across architectures, its rate does not — while LayerNorm removes the dependence entirely. Truong (2026) then show that under cross-entropy this runs through logit scale rather than the norm itself, but that channel is *absent* under MSE, where about half the effect survives unexplained. Kim (2026) instead make weight decay the clock, $t_{\text{grok}} \propto (1 - \beta)/(\eta\lambda)$. Weight norm, logit scale and λ are all scale-like or optimiser-level: none measures rotation, none separates a rotating kernel from the decaying one of Lewkowycz–Gur-Ari. H supplies the missing variable; the (a, b) fit separates the three.

Why this regime. Our baseline is MSE with vanilla gradient descent: where the logit-scale mediator is absent, where the theorem above applies verbatim, and where coupled L_2 and decoupled weight decay coincide, so Kim’s distinction collapses. The unexplained half of the norm effect then has a named candidate.

Practical. A_t is computable online from a fixed probe set and, unlike the training loss, is not flat during memorisation, so a crossing before the test loss falls is a genuine progress measure (Nanda et al., 2023). **Novelty, by exclusion.** Already published: the NTK–grokking link, the laziness knob α , eNTK spectral tracking, the norm delay law, mediation under intervention, and the MLP-versus-attention comparison. New here: (1) as a *timing* variable; the (a, b) test discriminating width-driven rotation from the λ -clock; and a rotation account of the norm law in the MSE regime, where the published mediator does not exist.

3 Feasibility

Baseline, data, cost. The dataset is $(a + b) \bmod p$, generated in a few lines. The baseline follows Kumar et al. (2024, App. 8.3) verbatim, so Tier 0 is checkable against a published figure: one hidden layer, $N = 100$, input $2p$ (concatenated one-hots), output p , MSE, full-batch GD, $\eta = 100$, 90% of the p^2 pairs for training, $p = 23$. A cross-entropy arm at the baseline cell links our constants to Truong et al. (2026), whose results are CE. Runs take minutes on one GPU. **[TODO: Colab/Kaggle, or a UQ Bunya account?]**



Figure 1: Reproduced grokking at baseline: train/test MSE with S_t and A_t overlaid.

Measured quantities. Train/test MSE; S_t , R_t and A_t on the centred kernel, $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$; relative parameter movement; and $y^\top K^{-1}y$, which predicts when linearisation breaks. The eNTK uses a fixed 256-point probe set (`torch.func.jacrev+vmmap`, sum-over-logits, 256×256 : seconds per checkpoint).

Fixed in advance. (i) t_{grok} is pre-registered as the step gap between the training and test losses crossing τ_{train} and τ_{test} , with τ -sensitivity reported; cells that never grok are right-censored, not dropped, so the fits are not selection-biased. (ii) Laziness uses the centred, rescaled predictor $\tilde{f} = \alpha[f(x, \theta) - f(x, \theta_0)]$ with $\eta = \eta_0/\alpha^2$ (Chizat, Oyallon, and Bach, 2019); α , initialisation scale and label rescaling are one knob, so we sweep only α . (iii) Weight decay is swept in $\eta\lambda$, which carries the GD stability bound $\eta\lambda < 2$ ($\lambda < 0.02$ at $\eta = 100$).

Design and budget. An $\alpha \times \eta\lambda$ grid (6×6) at baseline width; a $N \times \eta\lambda$ factorial ($\{50, 100, 400, 1600\} \times \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$) for the (a, b) fit; a clamp arm $\rho \in \{1, 1.25, 1.5, 1.75, 2\}$; and $p \in \{23, 29, 31\}$, training fraction $\in \{50, 70, 90\}\%$. Five seeds per cell, ten on the clamp arm where the MSE effect is half-size: ≈ 350 runs, ~ 20 GPU-hours. **[TODO: time one run in week 1 and resize; does $p = 23$ (529 pairs) grok cleanly?]**

Work plan. Five tiers, each independently reportable. **0:** reproduce the published α

sweep — a hard gate in week 1, since failure there makes downstream work uninterpretable. **1**: measure $S_t, R_t, A_t, t_{\text{grok}}$ over the $\alpha \times \eta\lambda$ grid; test the invariance of A_* and exhibit a cell where relative kernel movement is large while A_t is flat. **2**: the (a, b) fit, with bootstrap intervals and the residuals where a power law fails — the head claim. **3** (stretch): replicate the clamped-norm law under MSE and test whether A_* is invariant to ρ . **4** (theory, droppable): bound $\|T_t\|$ for the two-layer quadratic-activation network (making $k = 3$), whose grokked solution is closed-form (Gromov, 2023), giving analytic A_∞ and $t_{\text{grok}} \sim A_\infty/\dot{A}$; Eq. S5 is exact, so Tiers 1–2 stand regardless. **Division of labour** **[TODO: this is total placeholder]**: infrastructure, Zac Kienzie; Tier 0 reproduction, Aaron D’Mello; kernel metrics, Jasper Chong; sweeps and fitting, Samuel Rieger; theory and write-up, Zain Al-Saffi.

References

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Draft notes — delete this page before submission

Body is exactly two pages; references are excluded, per the brief. All citations were checked against primary sources on 31 Aug 2026.

1. **What changed in this revision.** The hypothesis is now a *threshold* claim (A_* invariant, $t_{\text{grok}} \perp S_t \mid A_t$) plus a *rate-law* claim tested by estimating (a, b) in $\log t_{\text{grok}} = c + a \log N + b \log(\eta\lambda)$. That regression, not the old mediation test, is the head claim: it is the one experiment that separates the rotation clock, the scale clock and Kim’s weight-decay clock, and it fails informatively (both exponents nonzero $\Rightarrow$ two clocks). Tier 3 was demoted to a stretch replication because Truong (2026) already ran a norm mediation test under cross-entropy.
2. **Verified numbers** in Truong et al. (2026): $\|W\|_c = \{42, 48, 55, 61, 67\}$ for $p = \{29, 41, 59, 79, 97\}$, $\|W\|_c \propto p^{0.38}$; $t_{\text{grok}} \propto e^{\alpha\rho}$ with $\alpha \approx 7.4\text{--}7.5$ (MLP) against ≈ 15 (unnormalised attention), $R^2 > 0.99$; “the form of the delay law is cross-architectural while its rate is not”; LayerNorm removes the dependence. Their loss is cross-entropy — hence our CE arm. Truong (2026): logit-scale mediation $R^2 = 0.97$ under CE, channel absent under MSE, norm effect $\approx$ half size.
3. **Verified as stated:** Lewkowycz & Gur-Ari Thm. 1 and Thm. 2 (eigenvalues decay, eigenvectors static; assumes MSE, $k \geq 2$, gradient flow, infinite width, Dyer–Gur-Ari conjecture), Eq. (S5) with T_t from (S6); Kumar App. 8.3 config and App. 14 (not 13); Mohamadi et al. (ICML 2024, PMLR 235); Atanasov et al.; Gromov’s quadratic-activation closed form; Cortes et al., JMLR 13, 795–828. **Not verified:** venue lines for Jacot (NeurIPS 2018), Nanda (ICLR 2023), Power (ICLR MATH-AI) — ids and authors are confirmed, venues are from memory.
4. **Open decisions.** Coordinator sign-off (scope email to Dr Nan Ye before 14 Sep); AI-use statement here or in the report; whether “pre-registered” means public (OSF) or fixed in the repo before sweeps run; whether the CE arm is one cell or a full $\alpha \times \eta\lambda$ replication; whether to keep the p and training-fraction arms at all, given they buy generality rather than discrimination.
5. **Held for the 12-page report:** the derivation of (1); the tier table; uncentred-alignment and first-logit eNTK robustness variants; the censoring model; $y^\top K^{-1}y = \|w - w_0\|^2$ near the kernel limit; the confound demonstration figure. An abstract would cost about six lines here.
6. **Template.** Font size, page size and the “don’t change” lines untouched; added `array`, `titlesec` (tighter section spacing), tighter display skips. Reverting lines 39–44 overflows by roughly four lines. Submit as *G2.proposal.<SubmitterName>.pdf*.